INTRODUCTION TO AZURE APP SERVICES

MODULE: APPLICATION DEVELOPMENT 3A

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LEARNING OBJECTIVES

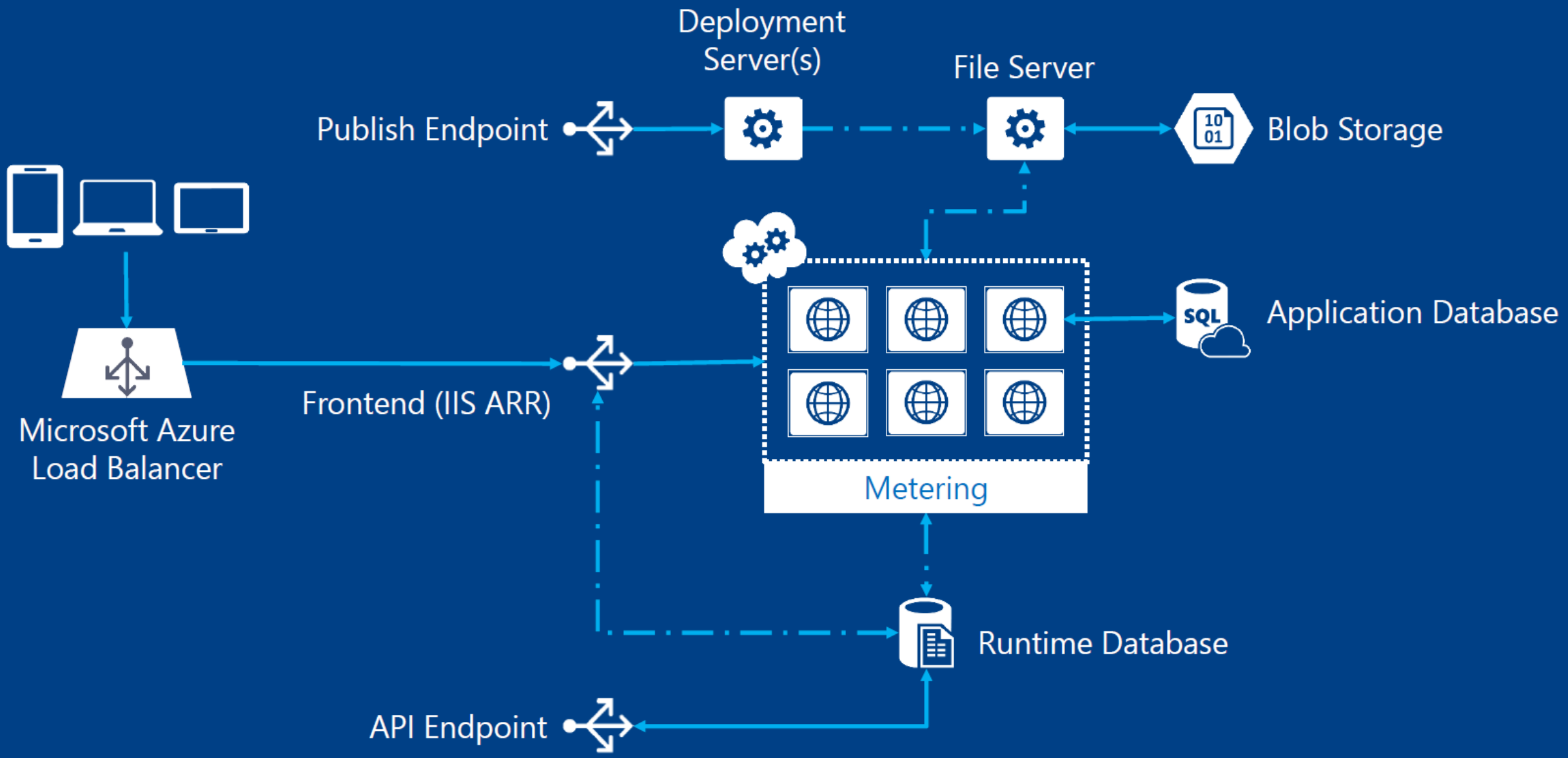
- Introduction to Azure App Services
  - What is an App Service?
  - App Service Web App Architecture
  - Supported Web Frameworks
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  - Scaling an App Service
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WHAT IS AN APP SERVICE?

- The App Service is a service that can host any of the following applications/API's:
  - ❑Web Apps
  - ❑Mobile Apps
  - ❑Logic Apps
  - ❑API Apps
  - ❑Function Apps
- Each app runs in its app service
- App Services support both Linux and Windows
- Web App for Containers hosted on the Linux platform

App Service Web App Architecture

Microsoft Azure



Microsoft

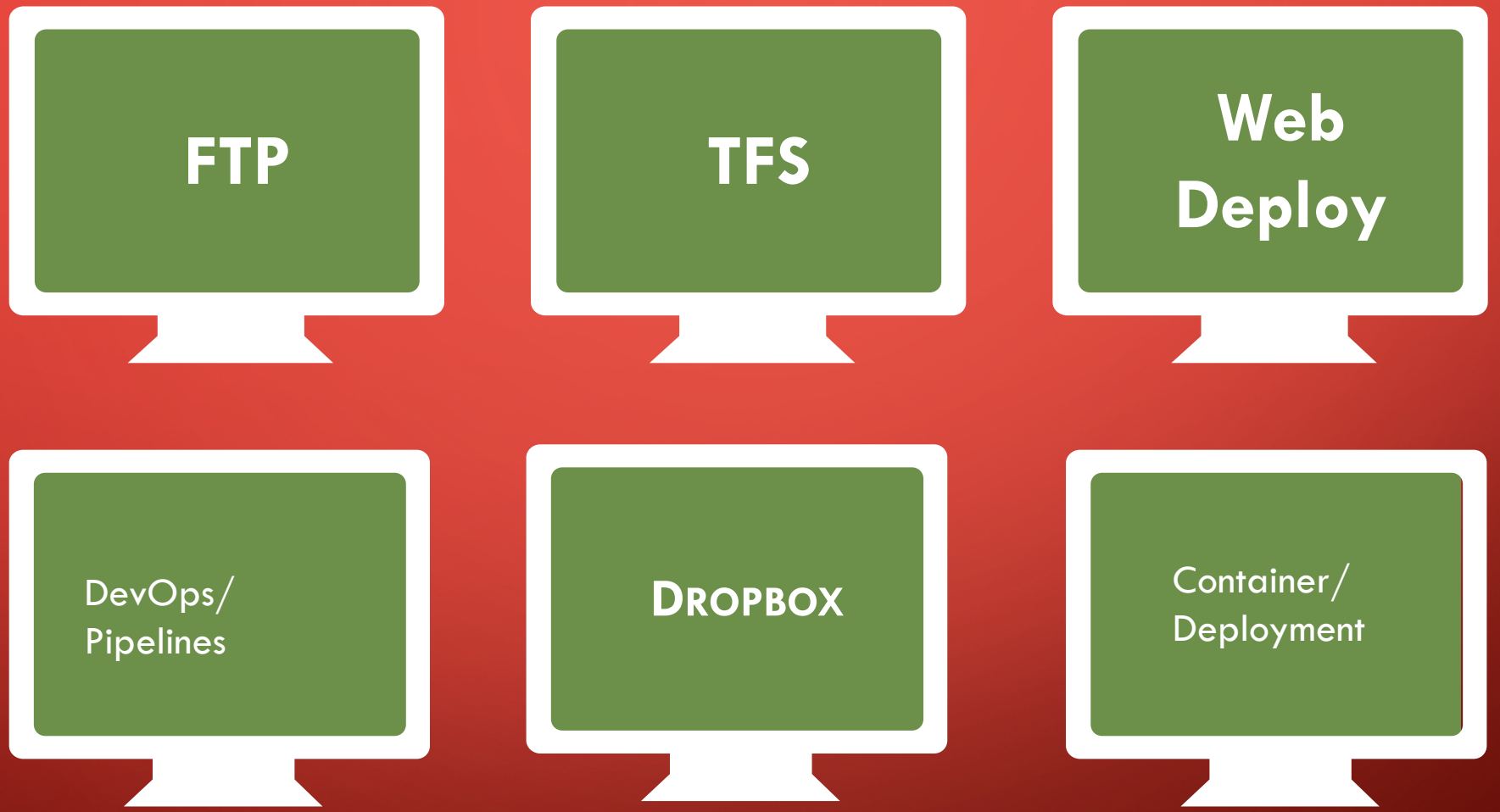
SUPPORTED WEB FRAMEWORKS



ASP.NET, ASP.NET Core, Java, Ruby, Node.js, PHP, Python, Docker Containers

Or any custom FastCGI Handler

SUPPORTED PUBLISHING METHODS



Azure Power Shell/Visual Studio

PUBLISHING METHODS

1. **Deployment from Visual Studio**:

   Developers can deploy their web applications directly from Visual Studio, Microsoft's integrated development environment (IDE). Visual Studio provides built-in tools and wizards for publishing web applications to Azure App Services, streamlining the deployment process.

2. **Azure PowerShell**:

   You can use Azure PowerShell to deploy web applications to Azure App Services by scripting the deployment process and executing PowerShell commands.

3. **FTP/SFTP Deployment**:

   Azure App Services supports FTP (File Transfer Protocol) and SFTP (Secure File Transfer Protocol) for manual deployment of web applications. You can upload your application files to the Azure App Service using FTP/SFTP clients such as FileZilla, WinSCP, or built-in browser-based FTP/SFTP clients provided by Azure.

4. **Azure DevOps Pipelines**:

   Azure DevOps Pipelines allow you to automate your web applications' build and deployment process. You can create, build, and release pipelines using Azure DevOps, configure triggers to automatically deploy changes to Azure App Services, and integrate with source control systems for continuous integration and deployment (CI/CD).

5. **Container Deployment**:

   Azure App Services supports deploying web applications as Docker containers. You can build your application into a container image, push it to a container registry such as Azure Container Registry or Docker Hub, and deploy it to Azure App Services using Docker Compose or Kubernetes.

WHAT IS AN APP SERVICE PLAN?

- An App Service plan defines the performance characteristics of the machine that will host the App Service.
- It defines the location for the App Service (such as West US)
- The Pricing tier (such as Free, Standard, or Premium) providing distinct settings for a variety of performance and service capabilities:
  - Number of cores or instance size
  - Amount of memory
  - Amount of storage
  - Maximum number of instances
  - Auto scaling options (depends on tier—automatic, manual, or none)
  - Resource Group (Container for Resources)
- The VM is abstracted away by Azure (No need to login).
- Multiple App Services can use the same plan or use separate plans.
- Keeping applications together allows them to scale together.

WHAT IS A WEB APPS?

- Web role is frontend of an application.
- Full capability set available including:
  - .NET, Node.js, Java, PHP, and Python
- Integrated VS publish, remote debug…
- CI with GitHub, BitBucket, VSO
- Auto-load balance, Autoscale, Geographical Disaster Recovery
- Virtual networking and hybrid connections
- Site slots for staged deployments

SCALING APP SERVICES

- App Services can be scaled up or out.
- Scale Up means doubling the resources.
- Scale Out means increasing the number of instances.
- The service plan can also be configured to scale dynamically/by time.

WHAT IS A WEB JOB?

- Web Jobs allows users to run a program or script in the same context as a web app.
- There is no additional cost to use Web Jobs.
- There are two types of Web Jobs (continuous or triggered).
- Continuous Web Jobs run immediately
- Continuous Web Jobs run on all instances (You can optionally restrict to one instance)
- Continuous Web Jobs typically do their work inside an endless loop.
- Trigger Web Jobs run manually or on a schedule.
- Triggered Web Jobs run on a single instance.
- Supports remote debugging.
- Supported file types:
  - .cmd, .bat, .exe (using Windows cmd), .ps1 (using PowerShell), .sh (using Bash), .php (using PHP), .py (using Python), .js (using Node.js), .jar (using Java)

SETUP AN APP SERVICE - DEMO

PUBLISHING TO AZURE FROM VISUAL STUDIO 2017

- In **Solution Explorer**, right-click the newly created Azure project, and, from the context menu, select **Publish**.
- Choose Microsoft Azure App Service
- Select an account or select **Add an account** in the account dropdown list (Azure Account).
- Give the App a name
- Choose the subscription to use for your deployment.
- Choose/Add a resource group
- Choose/Add a service plan
- Visual Studio creates a publishing profile for the application
- Click Create

DEPLOY FROM VISUAL STUDIO- DEMO